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Paper Title:

An LLM-Based Approach for Clarifying the Decisions of Vision Models in Autonomous Vehicles

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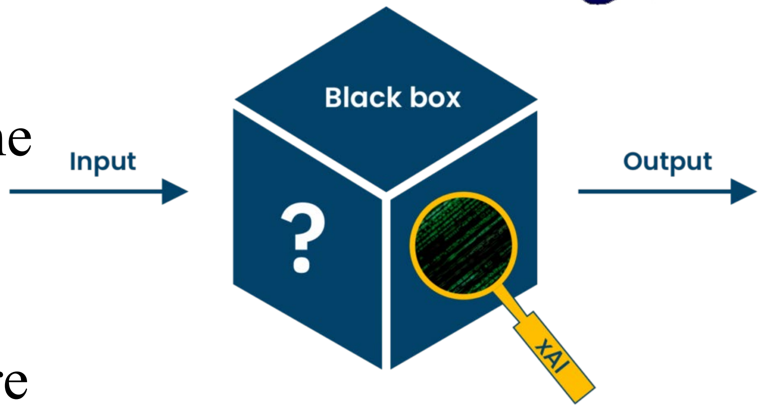
Problem statement:

Explainability

- Explainable Artificial Intelligence (XAI): A set of methods and techniques in machine learning that aim to explain the reasons for the decisions of complex models and reduce their "black box" nature.
- Conventional explainable methods and their limitations
- So, usually in more complex problems, textual explanations are more appropriate



Action: Slow down

Textual justification: Because the front vehicle has slowed down

Problem statement:

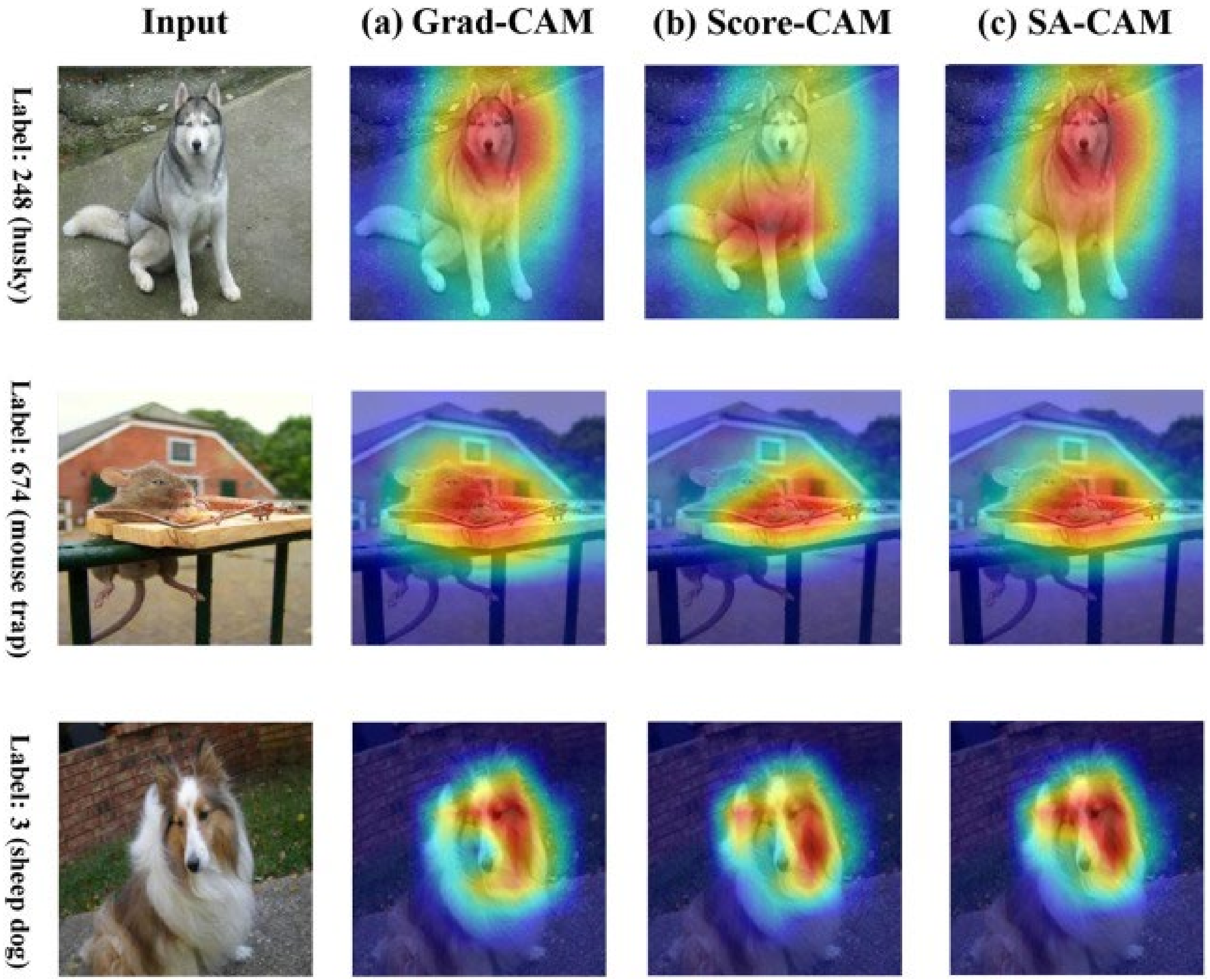
Use Case

Action description: The car is stopped **Action justification:** for the red light

Action description: The car pulls into the right lane
Action justification: because traffic is moving faster in that lane.

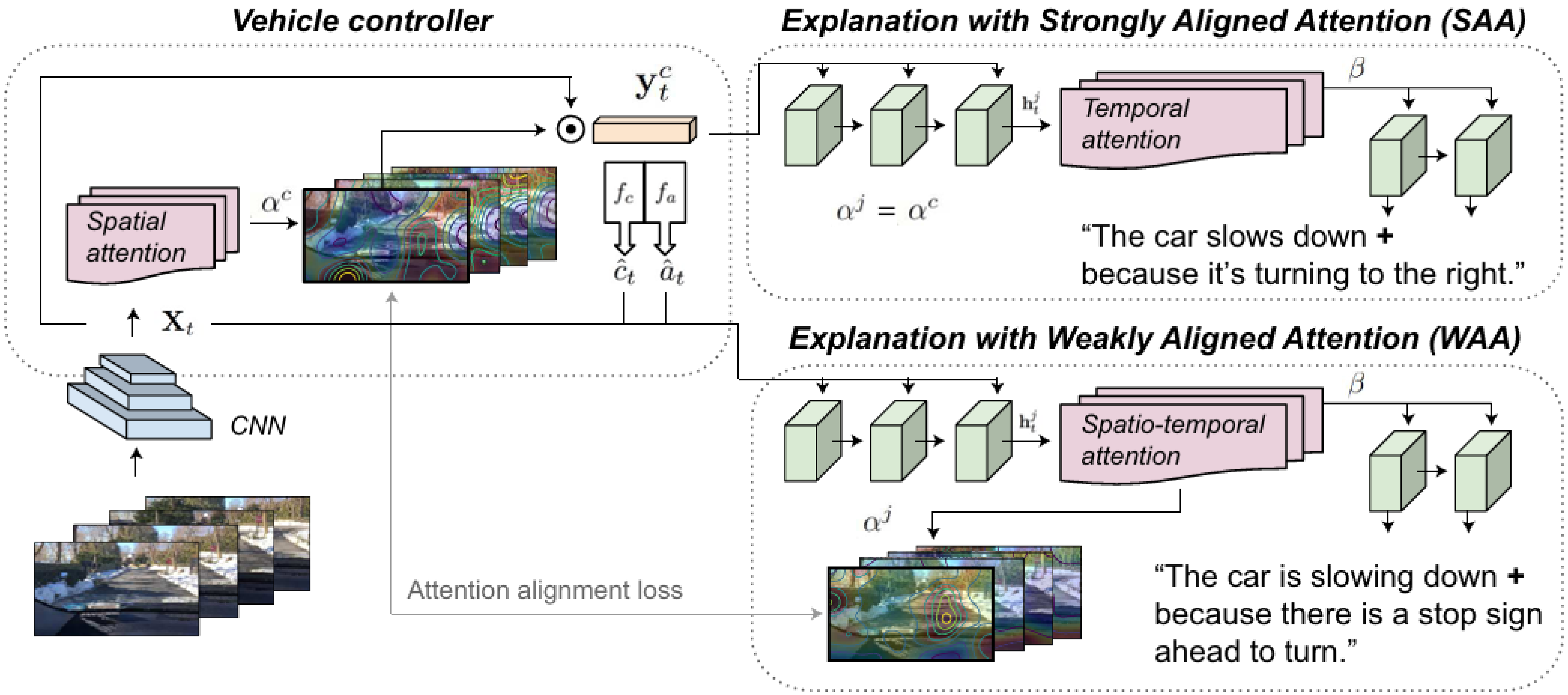
Problem statement:

Action Vs Justification



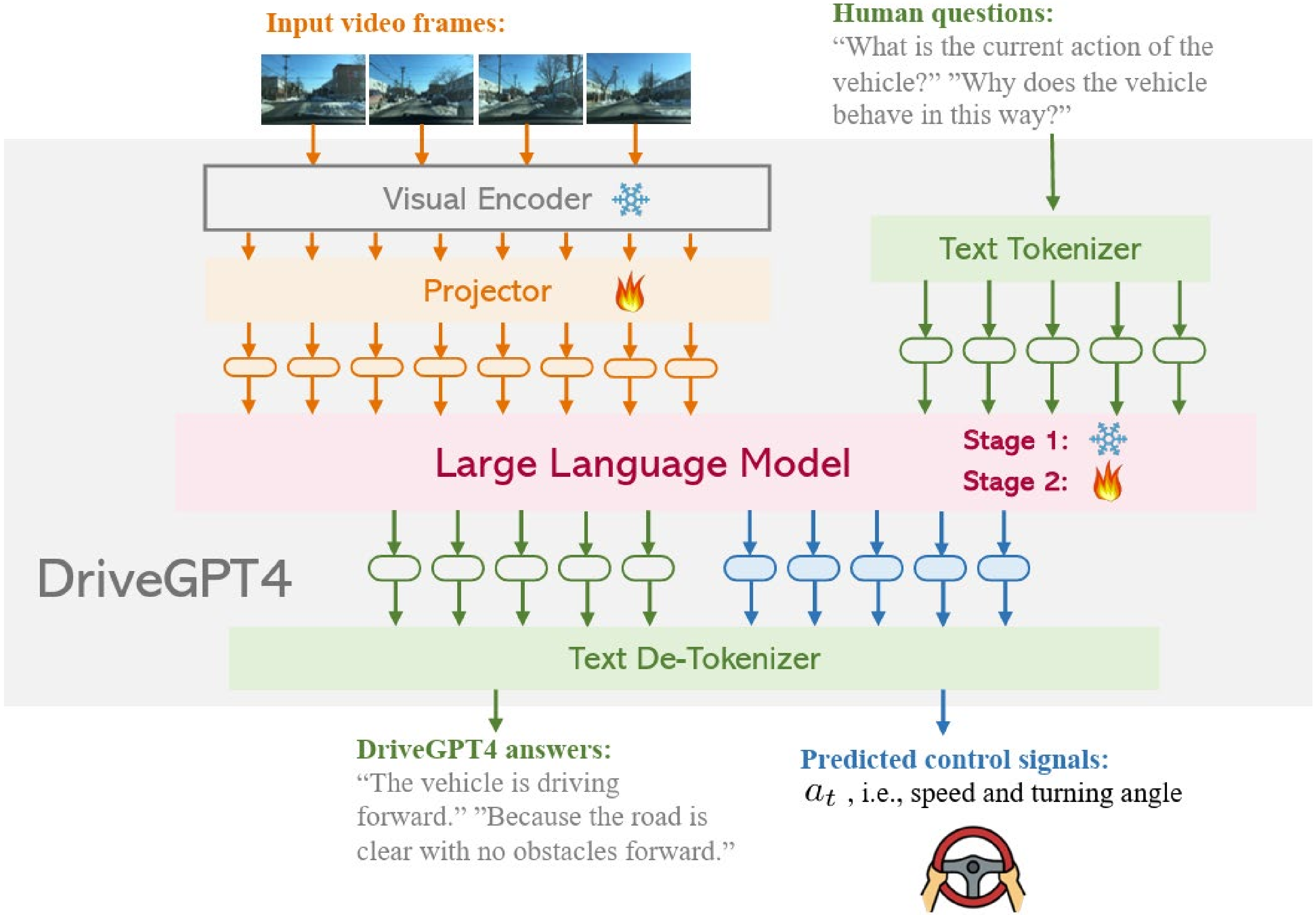
Previous methods:

Kim et al. (2018)



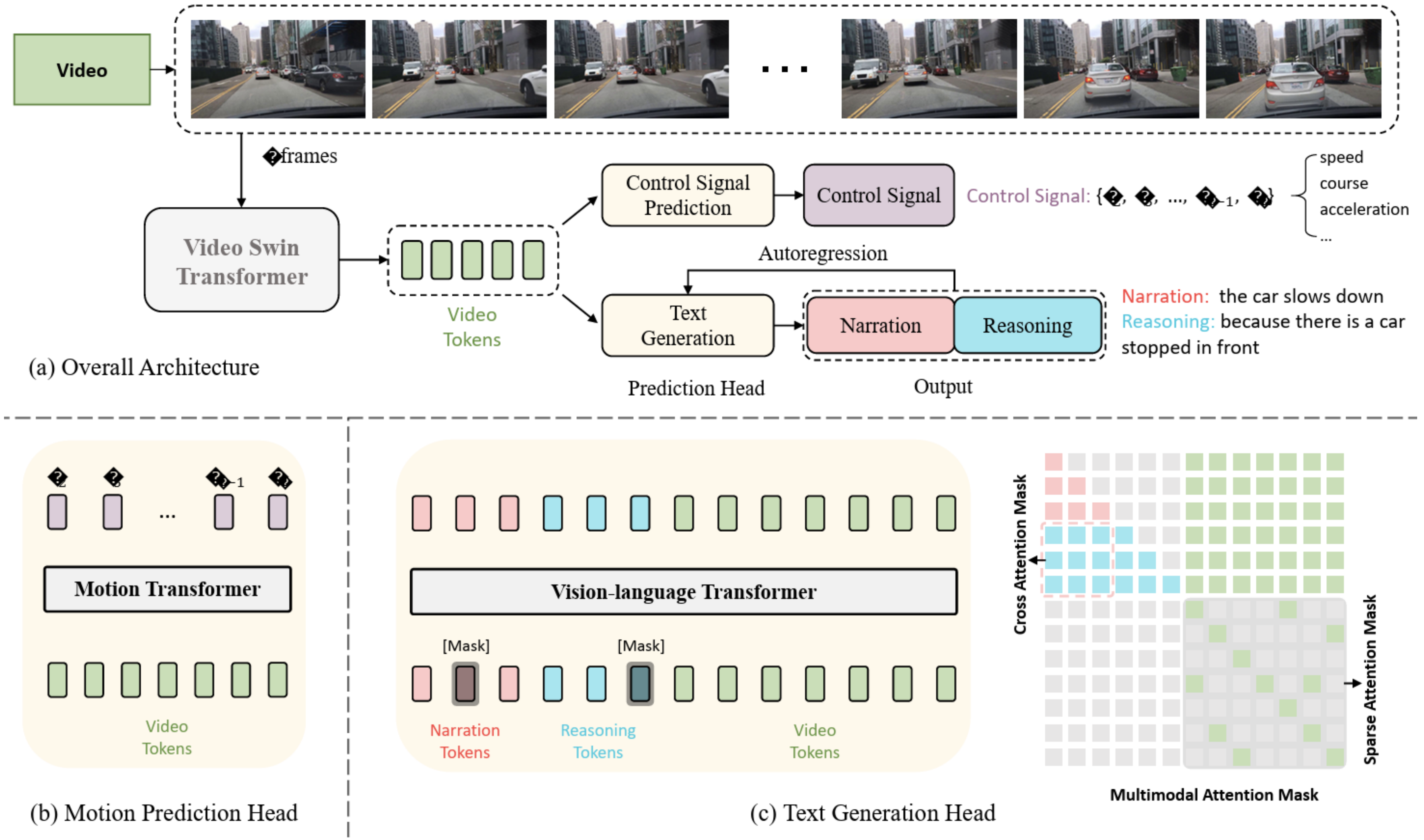
Previous methods:

DriveGPT4 (2024)

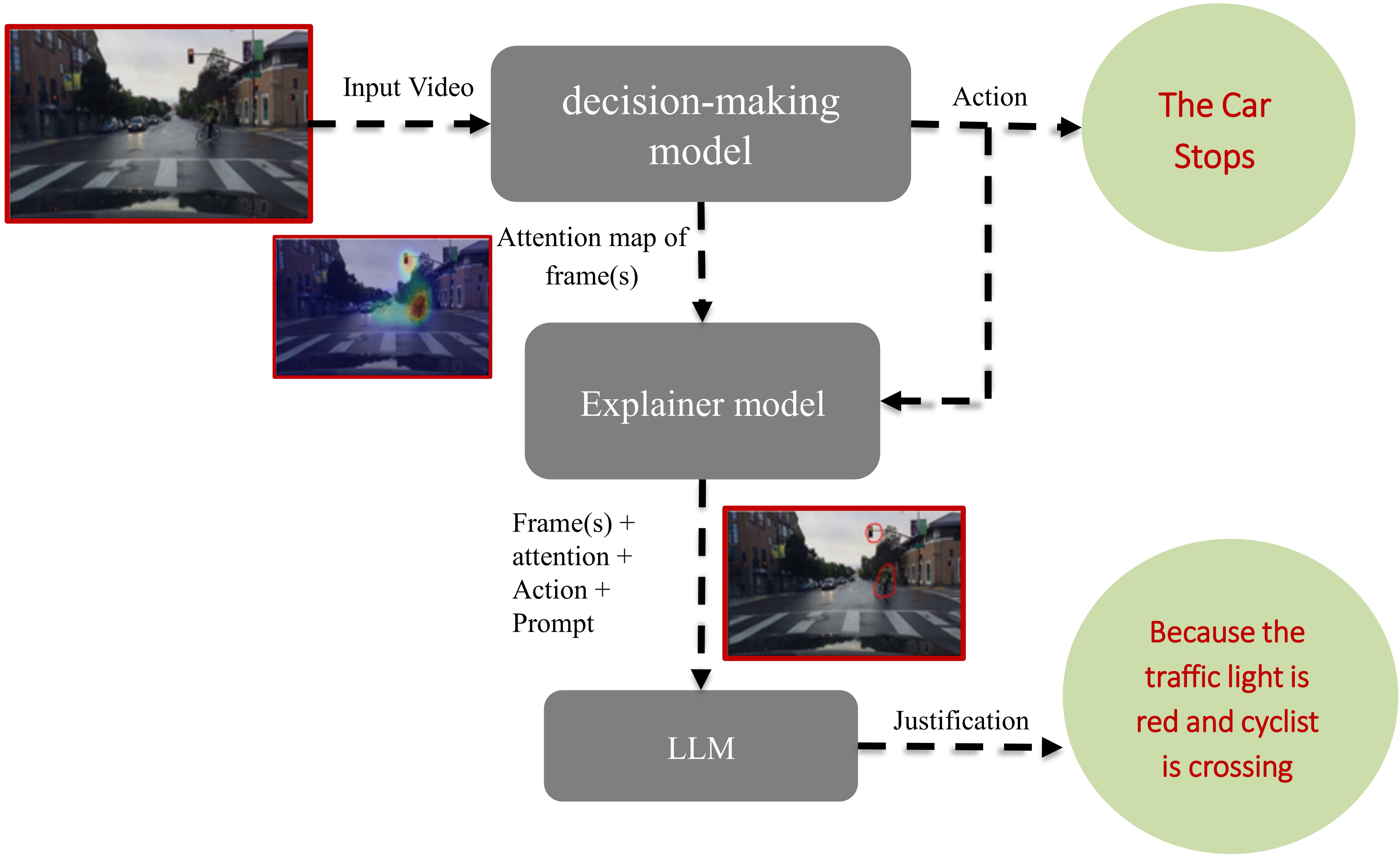


Previous methods:

ADAPT (2023)

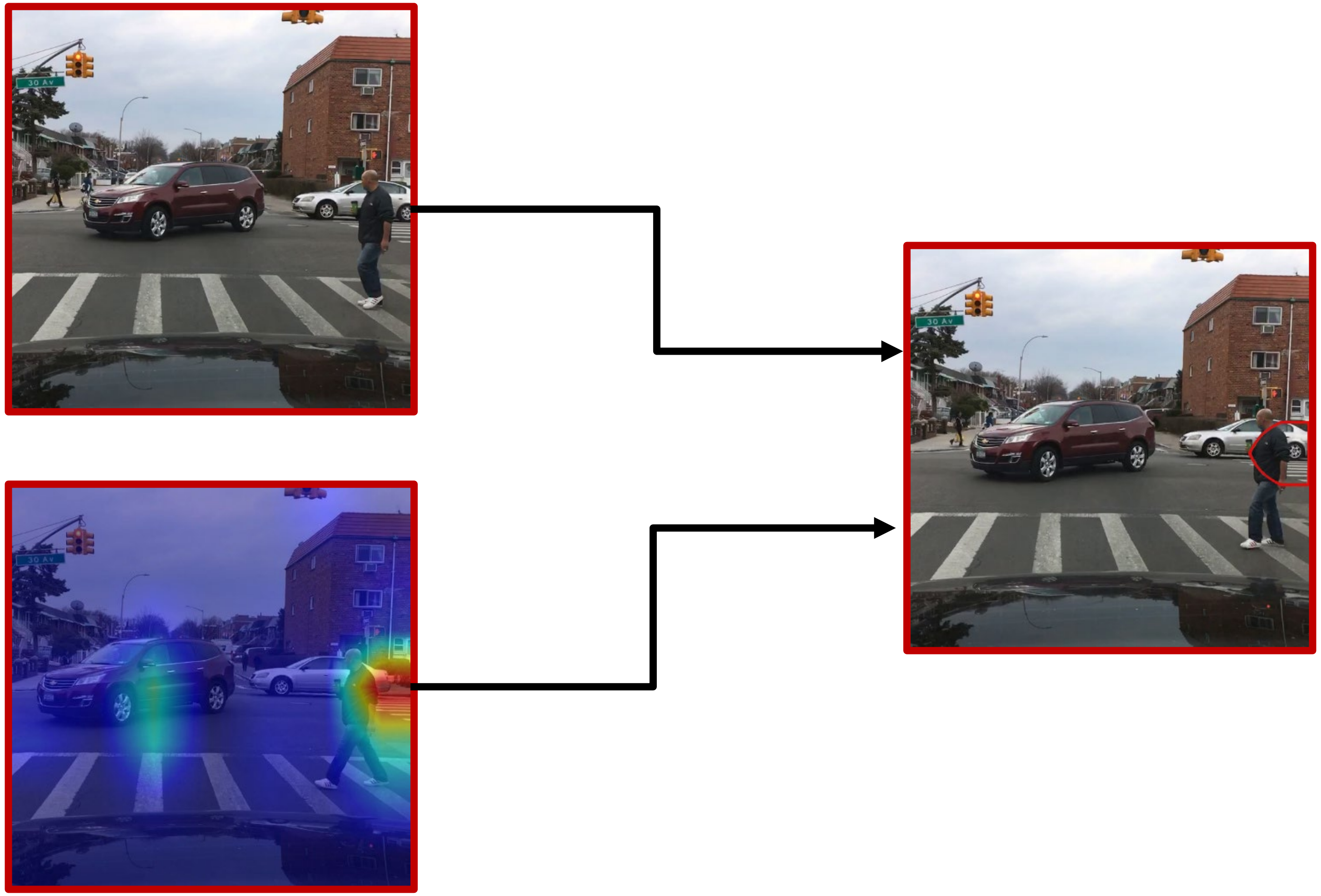


Proposed Method



Proposed Method

Attention map to red contours



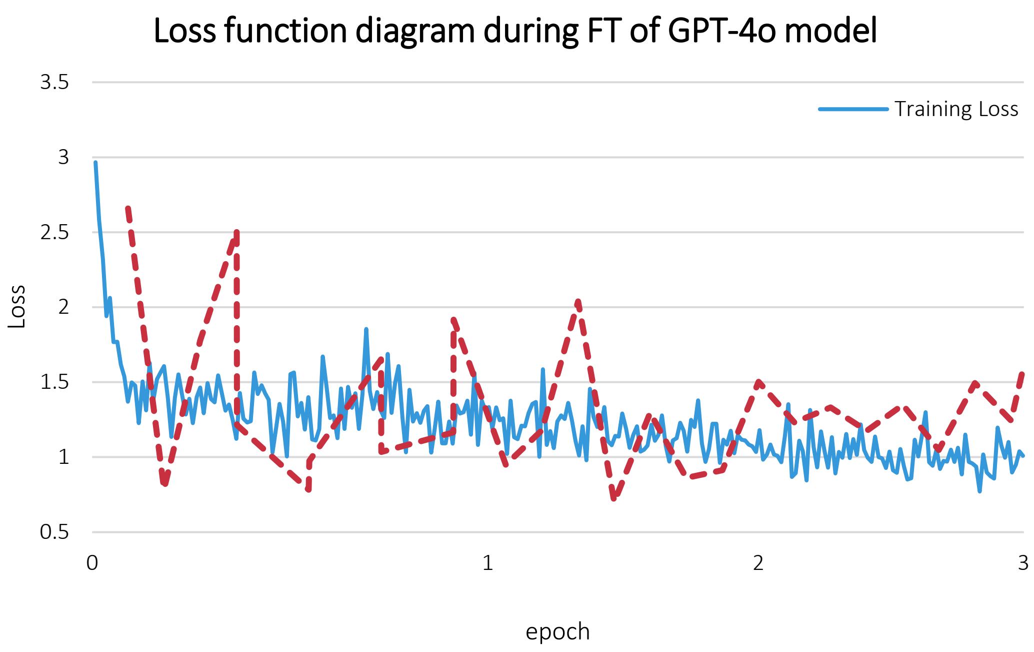
Proposed Method

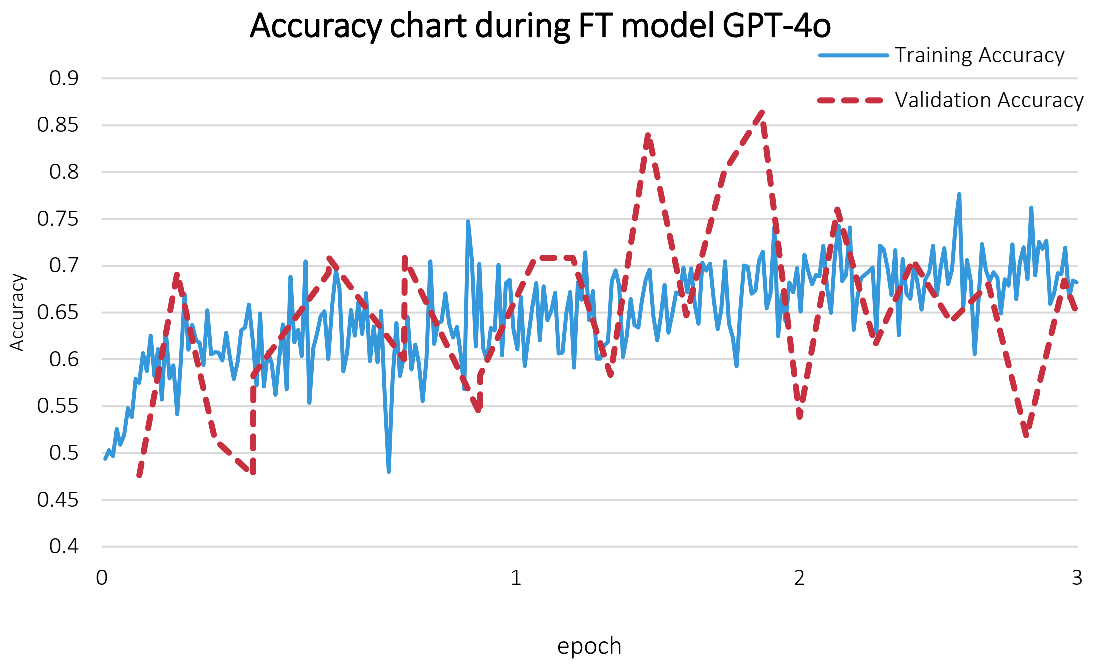
Prompt used in the LLM

Persona	You are a description system in driving scenarios.
Task	Your task is to capture an image of a driving scenario, the decision, and the area of interest of the decision-making system. Then explain why the decision-making system made that decision.
Input Data	Your input is {{frames_count}} frames of a camera in front of the vehicle, including the area of interest marked with a red circle.
Output Format/Style	Explain simply, briefly, and without explanation in only 1 or 2 short sentences. Avoid additional sentences, requests for more information, etc., do not explain context or given input information.
Context	You do not have access to the full information that the decision-making system does. So you consider that decision correct.
Constraints	Under no circumstances should you write more than 2 sentences. Avoid writing sentences like 'Based on the provided image ...' or 'In conclusion ...'.
Exemplars	Here are some examples: • since the truck in front wasn't moving fast. • because the light is green. • due to a vehicle in front of it moving slowly as it turns right.

Explainer Model Training and Review

Fine-Tuning on the GPT-4o Model

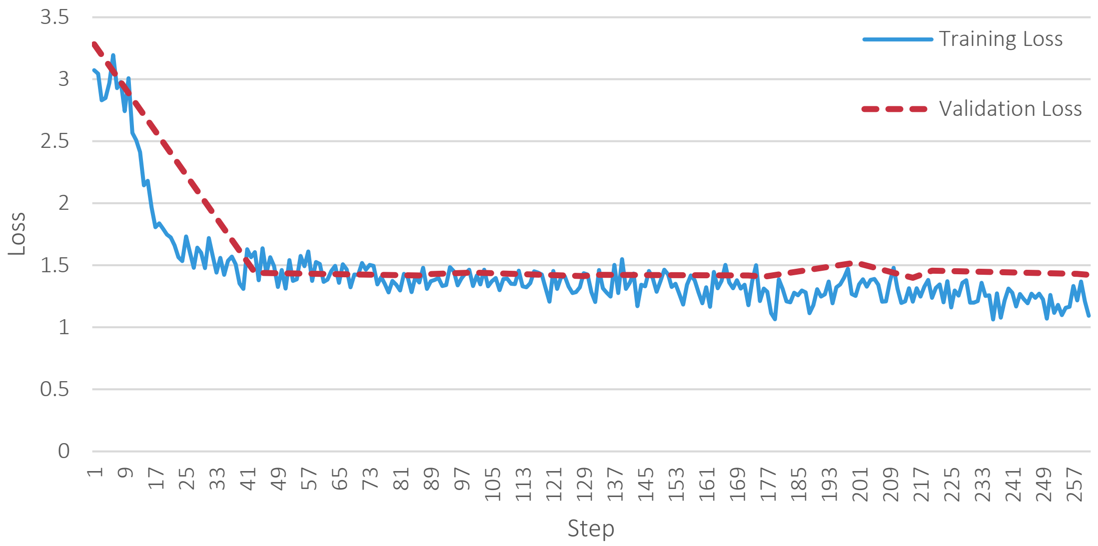




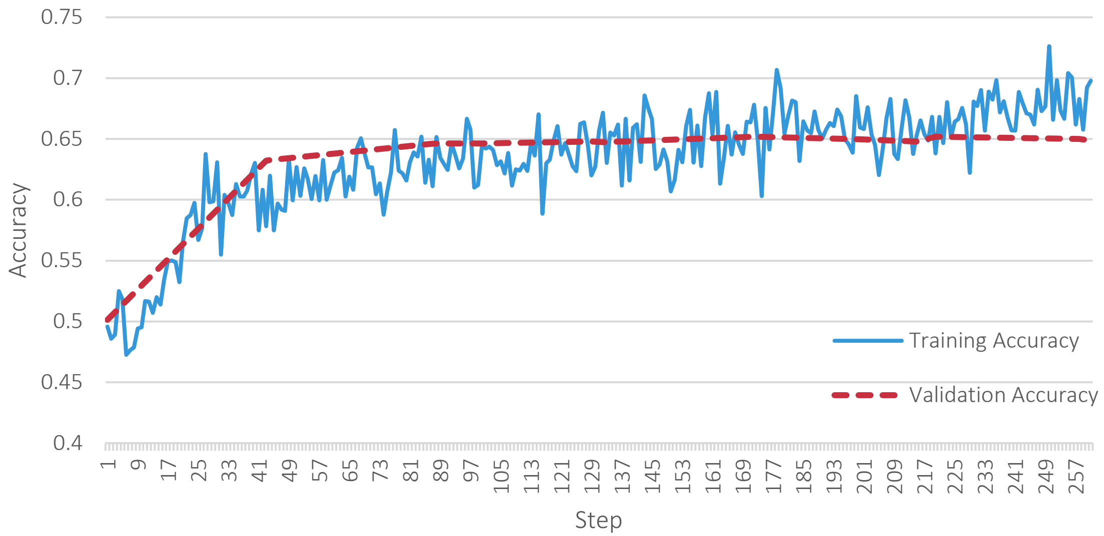
Explainer Model Training and Review

Fine-Tuning on the Gemini 2.5 Pro Model

Loss function diagram during FT of Gemini 2.5 Pro model



Accuracy chart during FT model Gemini 2.5 Pro



Results

	SPICE	ROUGE-L	METEOR	CIDER-D	BLEU-4
ADAPT	25.6	32	15.2	102.6	11.4
DriveGPT4		31.52		98.71	10.02
Gemini 2.5 Pro	24.35	32.82	17.72	62.2	7.08
Gemini 2.5 Flash	20.06	31.75	16.84	54.4	5.64
Gemini 2.5 Flash Lite	20.51	29.81	15.33	60.38	6.88
GPT-5 mini	11.4	23.29	14.05	25.48	2.03
FT-Gemini 2.5 Pro	**28.47**	**40.57**	**19.45**	**114.73**	**11.83**
FT-GPT-4o	26.25	33.84	15.24	103	11.09

Questions?

Thank you for your attention!